9 The Sun is one of many billions of stars in the Milky Way. The Sun emits a very large quantity of energy as electromagnetic radiation.

(a) State the **three** regions of the electromagnetic spectrum in which the Sun emits the most energy.

1 .....

2 .....

3 .....
[2]

(b) Electromagnetic radiation from the Sun travels at a speed of 3.0×10^8 m/s. The radiation takes 500 s to reach the Earth.

Calculate the distance from the Sun to the Earth.

distance = ... [2]

(c) Approximately 4.6 billion years ago, the Sun formed from an interstellar cloud of gas and became a stable star.

(i) Describe and explain what happens as an interstellar cloud of gas forms a protostar.

.....

.....

..... [2]

(ii) Describe and explain what happens as a protostar becomes a stable star.

.....

.....

.....

..... [3]

[Total: 9]

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